Brando Miranda

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Google Scholar; Website; Stanford Profile; MIT Website

Research Interests

AI for formal mathematics — Lean 4 theorem proving, autoformalization, and end-to-end formal verification of software and mathematics; contamination-resistant evaluation of LLM mathematical reasoning; data-centric machine learning for foundation and frontier models; alternative architectures (e.g. energy-based models) toward Artificial General Intelligence (AGI).

Education

Ph.D. in Computer Science *2022-2026 (expected)*
Stanford University GPA: 4.045/4.0
Advisor: Prof. Sanmi Koyejo, Stanford Trustworthy AI Research Group (STAIR).

Master of Engineering in Electrical Engineering and Computer Science *2014-2016*
Massachusetts Institute of Technology GPA: 4.8/5.0
Advisor: Prof. Tomaso Poggio, Center for Brains, Minds and Machines.
Thesis Title: *Function Approximation with Deep Neural and Gaussian Networks.*

Bachelor of Science, Computer Science and Engineering *2010-2014*
Massachusetts Institute of Technology Minors: Mathematics, Music

Awards & Honors

- **ICML 2026 Silver Reviewer** (May 2026) — top reviewer recognition, signed by the ICML 2026 Program Chairs.
- **Oral Recommendation (Top 1)**, 2nd AI for Math Workshop @ ICML 2025 — for *VeriBench: End-to-End Formal Verification Benchmark for AI Code Generation in Lean 4* (Miranda et al.); reviewer recommendation "Accept (Oral, Top 1)".
- **Pear AI Researchers Circle** (July 2025) — selective AI-researcher network convened by Pear VC; invited following the final round (R3) of the Pear AI Researcher Grant program.
- **ICML Workshop on Trustworthy Multi-modal Foundation Models and AI Agents (TiFA) — Outstanding Paper Award** (July 2024) — for "Why Has Predicting Downstream Capabilities of Frontier AI Models with Scale Remained Elusive?"
- **NeurIPS Outstanding Main Track Paper Award** (December 2023) — top 0.4% of NeurIPS submissions; only 2 main-track papers selected. For "Are Emergent Abilities of Large Language Models a Mirage?" (Schaeffer, Miranda, Koyejo).
- **EDGE Scholar**, Stanford University (September 2022) — Stanford fellowship supporting first-generation / low-income PhD scholars.
- **Stanford School of Engineering Fellowship** (September 2022) — multi-year departmental fellowship for incoming engineering PhDs.
- **Honorable Mention**, Ford Foundation Predoctoral Fellowship (2020, 2021) — national fellowship recognizing PhD applicants with potential to diversify the U.S. professoriate.
- **Best Research Project Award**, UIUC graduate course CS 598 "Learning to Learn" (December 2020).
- **Most Cited Paper Certificate**, International Journal of Automation & Computing (IJAC, December 2019) — for "Why and when can deep- but not shallow-networks avoid the curse of dimensionality: a review".
- **Computer Science Excellence Saburo Muroga Endowed Fellow**, UIUC (2019-2020) — top departmental fellowship for incoming CS PhDs.
- **Sloan Scholar**, Alfred P. Sloan Foundation Minority Ph.D. (MPHD) Program (2018-2019) — national fellowship for underrepresented STEM PhD students.
- **Grainger Engineering SURGE Fellowship**, UIUC (2018-2019) — multi-year college-level fellowship for diverse engineering PhDs.

Publications

Refereed Publications

(1) *VeriBench-FTP: A Formal Theorem Proving Benchmark in Lean 4 for Code Verification* (2025)
S. Barkallah, S. Daruru, Brando Miranda, Leni Aniva, Anima Nie, Sanmi Koyejo
5th Neural Information Processing Systems (NeurIPS) Workshop on Mathematical Reasoning and AI. 2025

(2) *VeriBench: End-to-End Formal Verification Benchmark for AI Code Generation in Lean 4* (2025)

Brando Miranda, Zhangir Zhou, Anima Nie, Elio Obbad, Leni Aniva, Kai Fronsdal, William Kirk, Dogus Soylu, et al.
2nd Workshop on AI for Math at International Conference on Machine Learning (ICML). 2025 (reviewer recommendation: **Accept (Oral, Top 1)**)
[OpenReview]

(3) *Pantograph: A machine-to-machine interaction interface for advanced theorem proving, high level reasoning, and data extraction in Lean 4* (2025)
Leni Aniva, Chuyue Sun, Brando Miranda, Clark Barrett, Sanmi Koyejo
International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS). 2025
[Springer Link]

(4) *Putnam-AXIOM: A Functional & Static Benchmark for Measuring Higher Level Mathematical Reasoning in LLMs* (2025)
Aryan Gulati, Brando Miranda, Eric Chen, Emily Xia, Kai Fronsdal, Bruno de Moraes Dumont, Sanmi Koyejo
International Conference on Machine Learning (ICML). 2025 & Neural Information Processing Systems (NeurIPS) Workshop on Mathematics and AI (MATH-AI). 2024
[arXiv]

(5) *Why Has Predicting Downstream Capabilities of Frontier AI Models with Scale Remained Elusive?* (2025)
Rylan Schaeffer, Hailey Schoelkopf, Brando Miranda, Gabriel Mukobi, Varun Madan, Adam Ibrahim, Herbie Bradley, Stella Biderman, Sanmi Koyejo
International Conference on Machine Learning (ICML). 2025 & International Conference on Machine Learning (ICML) Workshop on Trustworthy Multi-modal Foundation Models and AI Agents (TiFA) — Outstanding Paper Award. 2024
[arXiv]

(6) *CoDaPO: Confidence and Difficulty-Adaptive Policy Optimization for Post-Training Language Models* (2025)
Zhangir Zhou, Xingrun Lu, Cheng Cao, Brando Miranda, Tong Liu, Bo Han, Sanmi Koyejo
2nd Workshop on AI for Math at International Conference on Machine Learning (ICML). 2025
[OpenReview]

(7) *Causally Quantifying the Effect of Test Set Contamination on Generative Benchmarks* (2025)
Rylan Schaeffer, Brando Miranda, Jiaqi Kazdan, Kaivu Liu, Amir M. Ahmed, Niloofar Mireshghallah, et al.
Neural Information Processing Systems (NeurIPS) Workshop on Evaluating the Evolving LLM Lifecycle. 2025

(8) *Position: Machine Learning Conferences Should Establish a 'Refutations and Critiques' Track* (2025)
Rylan Schaeffer, Jiaqi Kazdan, Yarin Denisov-Blanch, Brando Miranda, Max Gerstgrasser, et al.
Neural Information Processing Systems (NeurIPS), Main Track. 2025
[arXiv]

(9) *Failures to Find Transferable Image Jailbreaks Between Vision-Language Models* (2024)
Rylan Schaeffer, Dan Valentine, Luke Bailey, James Chua, et al., Brando Miranda, et al., Sanmi Koyejo, Ethan Perez
International Conference on Learning Representations (ICLR). 2024 & Neural Information Processing Systems (NeurIPS) Workshop on Red Teaming Generative AI, Contributed Talk. 2024
[OpenReview]

(10) *Does Maximizing Neural Regression Scores Teach Us About The Brain?* (2024)
Rylan Schaeffer, Mikail Khona, Sankarshan Chandra, Michael Ostrow, Brando Miranda, Sanmi Koyejo
Neural Information Processing Systems (NeurIPS) Workshop on Unifying Representations in Neural Models (UniReps), 2nd Edition. 2024
[OpenReview]

(11) *Quantifying the Importance of Data Alignment in Downstream Model Performance* (2024)
Keshav Chawla, Aditya Sahai, Marco DePavia, Sathvik Sundar, Brando Miranda
International Conference on Learning Representations (ICLR) Workshop on Data-Centric Machine Learning Research (DMLR). 2024
[arXiv]

(12) *An Evaluation Benchmark for Autoformalization in Lean4* (2024)
Aryan Gulati, Devanshu Ladsaria, Shubham Mishra, Jasdeep Sidhu, Brando Miranda

International Conference on Learning Representations (ICLR) Tiny Papers Track, Second Edition. 2024
[arXiv]

(13) *Are Emergent Abilities of Large Language Models a Mirage?* (2023)
Rylan Schaeffer, Brando Miranda, Sanmi Koyejo
Neural Information Processing Systems (NeurIPS), Main Track. 2023 — Outstanding Paper Award & Oral
Featured in: New York Times, Quanta Magazine, Forbes, Stanford HAI; Cited in 2024 Economic Report of the President
[arXiv] [OpenReview]

(14) *Beyond Scale: the Diversity Coefficient as a Data Quality Metric Demonstrates LLMs are Pre-trained on Formally Diverse Data* (2023)
Brando Miranda, Alycia Lee*, Patrick Yu, Oluwasanmi Koyejo* (*equal contribution)
International Conference on Machine Learning (ICML) Workshop on Data-Centric Machine Learning & International Conference on Machine Learning (ICML) Workshop on Deployable Generative AI. 2023
[arXiv]

(15) *Why and when can deep-but not shallow-networks avoid the curse of dimensionality: a review* (2017)
Tomaso Poggio, Hrushikesh Mhaskar, Lorenzo Rosasco, Brando Miranda, Qianli Liao
International Journal of Automation and Computing 2017, Most Cited Paper Certificate
[Journal]

Preprints and Technical Reports

(1) *AI Coding Benchmarks Need Proofs, Not Just Tests* (2026)
D. Amrollahi, M. Karimi, Brando Miranda, Leni Aniva, Chuyue Sun, Clark Barrett, Sanmi Koyejo
Preprint 2026 — Lean 4 formal-proof-based evaluation of AI-generated code
[PDF]

(2) *Lean-ing on Quality: How High-Quality Data Beats Diverse Multilingual Data in Autoformalization* (2025)
W. Chan, M. Souliman, J. Nordhagen, Brando Miranda, Elyas Obbad, Sanmi Koyejo
Preprint, arXiv:2502.15795
[arXiv]

(3) *ZIP-FIT: Embedding-Free Data Selection via Compression-Based Alignment* (2024)
Elyas Obbad, Iddah Mlauzi, Brando Miranda, Rylan Schaeffer, Kamal Obbad, Suhana Bedi, Sanmi Koyejo
Preprint, arXiv:2410.18194
[arXiv] [Code]

Professional Experience

Stanford University - Stanford, CA — *September 2022 - 2026 (expected)*
Ph.D. Student in Computer Science. Advisor: Professor Sanmi Koyejo

Amazon Web Services (AWS) - Cupertino, CA — *June 2024 - September 2024*
Applied Scientist Intern

Morph Labs - Remote — *October 2023 - December 2023*
Machine Learning Research Scientist Consultant

Wise Agents - Stanford Spin-out — *2023*
AI Research Consultant

IBM Research - Yorktown Heights, NY — *May 2022 - August 2022*
Graduate Research Intern

University of Illinois Urbana-Champaign - Urbana-Champaign, IL — *September 2018 - May 2022*
Ph.D. Student in Computer Science. Advisor: Professor Sanmi Koyejo

IBM Research - Yorktown Heights, NY — *May 2021 - August 2021*
Graduate Research Intern

MIT CBMM (Center for Brain Minds & Machines) - Cambridge, MA — *June 2015 - September 2018*

Research Assistant. Advisor: Professor Tomaso Poggio

Media Coverage

- **Hacker News / Y Combinator (2025)**: Putnam-AXIOM benchmark posted on Hacker News — our *Putnam-AXIOM* benchmark for higher-level LLM mathematical reasoning drew front-page attention on Y Combinator's tech-news forum
- **White House Economic Report of the President (2024)**: Our work on emergent abilities was cited in the 2024 Economic Report of the President – the White House's annual flagship document on national economic policy
- **Quanta Magazine (2024)**: "How Quickly Do Large Language Models Learn Unexpected Skills?"
- **Andrew Ng (2024)**: Endorsed our paper as evidence that AGI development will be smooth and predictable
- **The New York Times (2023)**: "Silicon Valley Confronts the Idea That the 'Singularity' Is Here"
- **Forbes (2023)**: "AI 'Emergent Abilities' Are A Mirage, Says AI Researcher"
- **AK / @_akhaliq on X (2023)**: AK shared our *Beyond Scale* paper on X — the influential AI-research-paper aggregator (Gradio founder; now at HuggingFace) tweeted our *Beyond Scale* paper to his large research audience (68.9K views)
- **Stanford AI Lab Blog (ICML 2023)**: Our *Beyond Scale* and *Is Pre-training Truly Better Than Meta-Learning?* papers were featured in the Stanford AI Lab blog roundup of ICML 2023
- **Additional coverage**: Stanford HAI, Y Combinator News, Vice, Medium, HackerNews, NeurIPS blog, Reddit

Invited & Contributed Talks

- **Applications of Trustworthy Machine Reasoning with AI Coding Agents** – AAAI 2026 Tutorial "Trustworthy Machine Reasoning with Foundation Models" (Part IV, 50 min), Singapore, January 2026
- **Emergent Abilities in Large Language Models: Mirage or an Elusive Predictive Frontier?** – Hong Kong Baptist University, Department of Computer Science Seminar, Hong Kong, January 2026
- **Failures to Find Transferable Image Jailbreaks Between Vision-Language Models** – NeurIPS Red Teaming GenAI Workshop, Contributed Talk, December 2024
- **Are Emergent Abilities of Large Language Models a Mirage?** – Stanford IEEE Invited Talk, Stanford, CA, 2023
- **The Curse of Low Task Diversity: On the Failure of Transfer Learning to Outperform MAML** – NeurIPS Meta-Learning Workshop, Contributed Talk, New Orleans, LA, December 2022

Teaching Experience

Stanford University *2023 - 2025*
Course Assistant (3 quarters) & Instructor of Record (Spring)

- CS 197 *How to Do CS Research* with Prof. Michael Bernstein — TA'd 3 quarters; **Instructor of Record** for the Spring offering
- Mentored undergraduate research that produced **2 workshop publications** (including ICML workshop tracks)

University of Illinois Urbana-Champaign *August 2020 - December 2020*
Graduate Teaching Assistant

- CS 446 Machine Learning
- Designed problem sets and exams; held weekly office hours

Massachusetts Institute of Technology *2014-2016*
Graduate Teaching Assistant

- Statistical Learning Theory & Applications (9.520/6.860) with Prof. Lorenzo Rosasco & Tomaso Poggio
- Introduction to Algorithms (6.006) with Prof. Nancy Lynch, Bruce Tidor and Aleksander Madry
- Design & Analysis of Algorithms (6.046) with Prof. Shafi Goldwasser, Dana Moshkovitz, Nir Shavit
- Introduction to Machine Learning (6.036) with Prof. Tommi Jaakkola, Suvrit Sra & Regina Barzilay

Leadership & Service

Research Mentorship *2016 - Present*

- Mentored undergraduate and graduate students at Stanford, UIUC, and MIT CBMM
- Led research teams on data quality metrics, meta-learning, and formal reasoning

Academic Service *2018 - Present*

- Founding member of Stanford AI for Lean, a research community advancing AI for Lean theorem proving and formalizing mathematics

- Reviewer for NeurIPS 2026, ICLR 2026, ICML 2026 (**Silver Reviewer** – top reviewer recognition), ICLR 2025, ICML 2025 AI4MATH Workshop, ICLR 2024 DMLR Workshop, NeurIPS 2023 MATH-AI Workshop, ICLR 2020, JMLR 2018
- Graduate advisor for Latinos in Computer Science (LCS) at UIUC
- Founded "Stanford Bachata Sensual & Brazilian Zouk" and "UIUC Bachata Sensual & Zouk" official student organizations

Outreach *2017 - Present*

- Distributed Research Experiences for Undergraduates (DREU) UIUC 2019
- Undergraduate Research Opportunity Program (UROP) MIT 2017-2018
- Engineering of Intelligence Team (EIT) CBMM MIT 2017-2018

Skills & Tools

Programming: Python, PyTorch, TensorFlow, JAX, Lean 4, OCaml, Coq

ML/AI Tools: Hugging Face, Weights & Biases, DSPy, git, UNIX

Languages: English, Spanish (native fluency)